

5.4 Research Recommendations

- Results of this assessment are sensitive to natural mortality because it is highly correlated with the scale of the population (i.e., the R_0 parameter). For this assessment, the range and age-dependence of natural mortality was estimated by an indirect method (Hamel and Cope 2022) and within BAM. Mark-recapture approaches (conventional, telemetry, close-kin) might make it possible to obtain direct estimates of natural mortality specific to black sea bass in the South Atlantic region. Some tag-recapture studies have demonstrated relatively high tag return rates for black sea bass, at least compared to those of other reef fishes of the southeast U.S.
- Further research on best methods and approaches to combine multiple estimates of discard mortality into a single population wide value would be beneficial. Additional experiments to estimate discard mortality in greater ranges of depths, locations, angler experience related to barotrauma mitigation techniques (venting and descender devices), and times of year to match fishing within the South Atlantic would better characterize the uncertainty in this critical parameter.
- More research is needed on the cause(s) of low recruitment in several South Atlantic reef-fish stocks, including black sea bass. This topic is currently being investigated by the SEFSC.
- The number of fish discarded by the commercial sector appears to be highly uncertain as seen by the shift in scale of estimates from SEDAR 56. Direct observation of fishing practices by observers or video are needed to determine reliable estimates of these sources of mortality to fish in the Southeast Atlantic Ocean.
- Establish a more comprehensive sampling program for ages and lengths of fish captured by the recreational fleet in all regions of the South Atlantic.
- Investigate the potential impact of sampling of age composition data from commercial catch by size class (small, medium, or large) on compositions used within the assessment model and if methods are needed to correct these data.
- Gather more depth data and discarding behavior (venting or descending device) from private boat anglers.
- The following are from SEDAR 56, and are still needed:

The assessment panel recommended increasing the number of age samples collected from the general recreational sector.

Black sea bass in the southeast U.S. were modeled in this assessment as a unit stock, as recommended by the DW and supported by genetic analysis (SEDAR 76-RD42). For any stock, variation in exploitation and life-history characteristics might be expected at finer geographic scales. Modeling such sub-stock structure would require more data, such as information on the movements and migrations of adults and juveniles, as well

as spatial patterns of recruitment. Even when fine-scale spatial structure exists, incorporating it into a model may or may not lead to better assessment results (e.g., greater precision, less bias). Spatial structure in a black sea bass assessment model might range from the very broad (e.g., a single Atlantic stock) to the very narrow (e.g., a connected network of meta-populations living on individual reefs). What is the optimal level of spatial structure to model in an assessment of snapper-grouper species such as black sea bass?

Protogynous life history: 1) Investigate possible effects of hermaphroditism on the [recent low recruitment]; 2) Investigate the sexual transition for temporal patterns, considering possible mechanistic explanations if any patterns are identified; 3) Investigate methods for incorporating the dynamics of sexual transition in assessment models.

In this assessment, the number of spawning events per mature female per year assumed a constant value of $X = 31$. That number was computed from the estimated spawning frequency and spawning season duration. If either of those characteristics depends on age or size, X would likely also depend on age or size. For black sea bass, does spawning frequency or spawning season duration (and therefore X) depend on age or size? Such dependence would have implications for estimating spawning potential as it relates to age structure in the stock assessment.

For this assessment, the age-dependent natural mortality rate was estimated by indirect methods. More direct methods, e.g. tag-recapture, might prove useful. Some tag-recapture studies have demonstrated relatively high tag return rates for black sea bass, at least compared to those of other reef fishes of the southeast U.S.